Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Software Engineer and data analyst with **8+ years building production systems and ML pipelines on regulated financial data**. Built **Isolation Forest anomaly detection** at GMU across 3 research pipelines - surfaced anomalies manual review missed, cut manual QA 70%. Applied Python pipelines across **1.6M financial records at Entrata** with output validation and quality gates; achieved **100% PHI compliance** on AWS S3 migration with zero security incidents. Expert in Python and SQL; advanced in scikit-learn, anomaly detection, and Tableau. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Anomaly Detection | Fraud Signal Detection | Python | SQL | Isolation Forest | Random Forest | Incident Response | Financial Data Pipelines | PHI / Compliance | Data Quality Gates | Tableau / Power BI | AWS S3 / EC2

EXPERIENCE

George Mason University
Aug 2024 - May 2026

Research Associate - ML Pipelines & Anomaly Detection

Fairfax, Virginia

- Built **Isolation Forest anomaly detection system** in Python across 3 active research pipelines - surfaced anomalies that manual review missed, **cut manual QA time by 70%**, and improved downstream model reliability.
- Built **regression models (R^2 = 0.977)** in Python and R across WHO and UK Road Safety datasets spanning **190+ countries** - multi-source data collection, transformation, validation, and stakeholder reporting.
- Designed **Tableau, ggplot2, and Quarto dashboards** for faculty leadership - improved research comprehension across two cohorts by **40%**.

Entrata India (Db Xento Systems)
Oct 2018 - Aug 2024

Software Engineer - Financial Data Pipelines & Compliance

Pune, India

- Applied Python and OpenAI API to automate data transformation pipelines across **1.6M financial records** - production systems on regulated data with output validation and quality gates.
- Achieved **100% PHI compliance** by migrating all client data to AWS S3 with Libsodium-encrypted pipelines - **zero security incidents**; built 5+ internal tools cutting resolution time by 40%.
- Cut deployment times by **40%** by building 20+ production microservices (Python, PostgreSQL, RabbitMQ) across 120+ Agile sprints - **100K+ MAU, 99.5% uptime** sustained over 6+ months.
- Improved production query performance by **30%** by tuning 200+ SQL queries across all client environments; containerized 30%+ of microservices with Docker, cutting execution time by 50%.
- Eliminated **80% of manual operational workload** via Python/PL-SQL automation; resolved 100+ high-priority bugs sustaining platform reliability across all production environments.

Entrata India (Db Xento Systems)
Jul 2017 - Sep 2018

Associate Software Engineer

Pune, India

- Improved system stability by **35%** building 20+ full-stack modules (Python, PostgreSQL) - 99.5% uptime and 25% faster report generation.

PROJECTS

Global Health & Road Safety Research - Anomaly Detection System

2024-2025 | Python, R, Isolation Forest, scikit-learn, Tableau, ggplot2, Quarto | George Mason University

- Built **AI-powered anomaly detection auditor (Isolation Forest)** across 3 active research projects - cut manual QA time by **70%** and improved downstream model reliability; the same pattern-recognition pipeline used in fraud signal detection.
- Built end-to-end data preprocessing pipelines and regression models (R^2 = 0.977) across **WHO + UK Road Safety datasets (190+ countries)** - multi-source data collection, transformation, output verification.

CrashClarity - Large-Scale Pattern Detection (104K Records)

2025 | Python, R, scikit-learn, Random Forest, Logistic Regression, Feature Engineering | Live: Portfolio

- Achieved **85%+ accuracy** on 104K+ records by building a Random Forest + Logistic Regression pipeline with feature engineering across multi-variable datasets - deployed interactive dashboard for non-technical stakeholders.

AI on FHIR - HIPAA-Compliant NLP Medical Records System

2024 | Flask, LLMs, FHIR, React, Docker, spaCy, BERT | GitHub

- Built HIPAA-compliant RBAC system with NLP query translation - reduced manual clinical retrieval time by **~60%** across structured medical records (same compliance discipline as financial fraud data).

EDUCATION

M.S. Data Analytics Engineering - George Mason University — 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, anomaly detection, ML systems, data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering — 2013-2017

TECHNICAL SKILLS

Languages: Python, SQL, R, JavaScript, Bash

ML & Detection: Isolation Forest, Random Forest, Logistic Regression, XGBoost, scikit-learn, Feature Engineering, Anomaly Detection, Predictive Modeling, NumPy, Pandas

Data & Infra: PostgreSQL, MySQL, AWS (S3, EC2, RDS, Glue), Docker, FastAPI, Flask, RabbitMQ, Microservices **Visualization:** Tableau, Power BI, ggplot2, Plotly, Quarto

Compliance: PHI/HIPAA compliance, Libsodium encryption, secure coding, audit logging, TDD/BDD **AI Tools:** LangChain, LangGraph, OpenAI API, BERT, spaCy, ChromaDB